

Introduction to PyRadiomics

1. GOAL

We will use the PyRadiomic library to extract a diverse range of features from medical images. This library offers a comprehensive set of tools that allow for efficient and accurate analysis of various aspects of medical imaging data.

2. MATERIALS

Session slides: Introduction to PyRadiomics,

Python code: PyCode_PyRadiomics.zip

Data: Full Dataset with VOIs.zip and MetadatabyNoduleMaxVoting.xlsx.

3. SESSION EXERCISES

Exercise 1. Feature extraction from a volume image and mask.

- a) Execute the code VolumeFeaturesExtraction.py and analyse its output. What are the features extracted? What is the meaning of each value associated to each feature?
- b) Identify each part/section of the output. Why could be useful to have this information.
- c) Make the corresponding modification to extract other type of features.

Exercise 2. Feature extraction from the slices of the image and mask.

- a) Execute the code SliceFeaturesExtraction.py and analyse its output. What are the features extracted?
- b) What are the main differences in the outputs generated by VolumeFeatureExtraction.py and SliceFeatureExtraction.py.

Exercise 3. Features extraction for all images and masks in the database.

- a) Observing the python code featuresExtraction.py, which files are used as input and output? What features are extracted? Where is the diagnosis included? Notice that this code applies pre-processing, can you explain what it does?
- b) Make the necessary modification to only extract all the GLCM features.
- c) Modify the featureExtraction.py file to extract features from all nodules in the database.